

Phase 2 of 4

Hospital–Doctor–Patient Connection System & Clinical Features

Maternal Health Platform — Development Roadmap · Derived strictly from the Patient, Doctor, and Hospital User Flow documents

What This Phase Covers

This phase builds everything that links a Hospital, its Doctors, and a Patient together — doctor registration by the hospital, the public doctor directory, the access-code mapping mechanism, and the clinical features (chat, appointments, video consultation, reports, prescriptions, medicine ordering) that unlock the moment a patient connects to a hospital. This phase assumes Phase 1's login/dashboard shells already exist.

1. Hospital Registers Its Doctors

Step 1 — Registering a New Doctor

The hospital enters the following for every doctor before any patient can be mapped to them:

- Doctor's name
- Qualification
- Visiting timings at this hospital
- Number of days per week the doctor visits
- Whether the doctor is Visiting or Permanent at this hospital
- Doctor's login email
- Password — set by the hospital or auto-generated as a temporary password

Flow: hospital clicks "Register New Doctor" → enters name, qualification, timings, days/week, permanent/visiting → enters doctor's email + sets/generates password → doctor profile is created and instantly visible in the public hospital directory → doctor can now log in with these credentials into THIS hospital's portal only.

NOTE: The credentials the hospital issues let the doctor log in only into this specific hospital's portal — not into the platform generally. This is distinct from the doctor's own personal account built in Phase 1.

Step 2 — Multi-Hospital Doctors — Independent Credentials Per Hospital

The same real-world doctor may visit more than one hospital. Each hospital registers him completely independently — separate profile, separate login email and password, separate patient list.

Hospital	Doctor Entry	Days/Timings	Login Credentials
Apollo Hospital	Dr. Ramesh — Permanent	Mon–Fri, 10am–4pm	ramesh@apollo portal — independent

Hospital	Doctor Entry	Days/Timings	Login Credentials
Fortis Hospital	Dr. Ramesh — Visiting	Saturday only, 9am–1pm	ramesh@fortis portal — independent

Step 3 — Editing and Deactivating a Doctor

Edits to timings, days, or status apply immediately across the doctor directory, patient profile view, and doctor's own profile. Deactivation is stricter:

- Doctor's login is disabled immediately
- Existing patients keep full historical access to chat history, reports, and prescriptions — read-only
- No new appointments can be booked with a deactivated doctor
- Connected patients see a banner: "Dr. [Name] is no longer available at this hospital"
- Hospital must reassign affected patients to an active doctor for continued care

Edge Cases — Doctor Registration

Edge Case	What Happens
Doctor registered with an email already used at this hospital	Blocked with a duplicate-email error.
Doctor registered with zero patients	Fully allowed — no minimum patient count to keep a doctor active.
Hospital tries to permanently delete a doctor with active patients	Hard delete is blocked. Only deactivation is allowed.
Doctor's visiting days change mid-week	Takes effect immediately for future slots; already-booked appointments unaffected.

2. Doctor Visibility Rules — The Public Directory

Every doctor's details — name, qualification, timings, days per week, permanent/visiting status — are visible in two places at once: the patient's profile and the doctor's own profile, symmetrically.

Who Is Viewing	What They See	Login Required?
Patient — not connected	Full doctor list of any hospital	No
Patient — connected	Same list, assigned doctor highlighted	No, already logged into the platform
Doctor — browsing other hospitals	Full doctor list of that hospital	No
Doctor — own hospital dashboard	Full doctor list including himself	Yes, into his own hospital's portal

NOTE: Browsing is not access. Public visibility of the doctor directory is read-only. It never grants login access to that hospital's portal, and no patient data is ever shown in this public view.

3. Hospital Maps a Patient to a Doctor — The Access Code

The hospital never creates the patient's login account — a patient always owns and creates her own account independently (Phase 1). What the hospital creates is a mapping record plus a one-time access code linking her existing (or soon-to-be-created) account to this hospital and doctor.

Step-by-Step Flow

- Hospital opens the doctor's profile → clicks "Add Patient"
- Enters patient's full name, phone number, and confirms assigned doctor
- System auto-generates an 8-character access code (e.g., BLOOM-7829) — valid 30 days, single use

- Hospital staff hands the code to the patient physically at her visit
- Patient enters the code on her own dashboard → "Connect to this Hospital"
- Her account is linked — chat, reports, appointments, and prescriptions unlock immediately

Doctor With Zero Patients — Adding Patients Later

A hospital can register a doctor with no patients yet. The hospital can click "Add Patient" under his profile at any time afterward — no limit, no minimum.

Edge Cases — Patient Registration and Mapping

Edge Case	What Happens
Staff forgot to hand the code, or patient lost it	Dashboard still shows the code as "Unused." Staff can view, reprint, or resend it within its 30-day window.
Same patient registered twice by mistake	Hospital searches by phone number before creating a new mapping; duplicate is reused/edited instead.
Patient needs to move from Doctor A to Doctor B	Handled by the Reassign action; no new access code needed.
Patient never uses the code within 30 days	Code auto-expires; hospital regenerates a fresh one, old code permanently invalidated.

4. Access Code Lifecycle

An 8-character alphanumeric code, format HOSPITALPREFIX-#### (e.g., BLOOM-7829), valid 30 days from generation, single-use — marked used the moment the patient enters it successfully.

Code Status on Dashboard	Meaning	Hospital Action Available
Active / Unused (green)	Generated, not yet entered by patient	Reprint, resend, or wait
Used (grey)	Patient successfully connected	None needed — mapping now visible under the doctor
Expired (red)	30 days passed, still unused	Regenerate — creates a new code, permanently invalidates the old one

Edge Cases — Access Code Lifecycle

Edge Case	What Happens
Patient enters a valid code on the wrong hospital's connect prompt	Blocked: "This code does not belong to this hospital." Code is scoped to a specific hospital-doctor pairing.
Hospital regenerates a code mid-submission by the patient	Old code invalidated instantly; in-flight submission fails with "Code no longer valid — request a new one."

5. Patient Side — Browsing and Connecting to a Hospital

Step 1 — Browsing Any Hospital — No Code Needed

From the Hospital Management Dashboard the patient searches any hospital name and sees its public page, without any connection required:

- Hospital name, full address, area, district, and pincode
- Complete list of all doctors registered in that hospital
- Each doctor's name, qualification, visiting timings, days per week, and permanent/visiting status

Step 2 — Connecting to a Hospital — The Access Code

Outcome	Condition	What Happens
Valid	Code correct + within 30 days + unused	Connection established immediately; code marked used; never needed again.
Wrong code	Incorrect characters entered	"Incorrect code. Please check and try again." Patient may retry.
Expired	30 days have passed since generation	"Code expired. Contact your hospital for a new code." Hospital regenerates.

Step 3 — After Successful Connection — What Unlocks Immediately

One code entry unlocks everything for that hospital simultaneously:

- Assigned doctor's full details visible in her dashboard
- Tele-consultation chat — direct human-to-human messaging with the doctor
- My Reports — medical reports uploaded by the hospital, viewable as PDF
- Book Appointment — see available slots from the doctor and book directly
- Video Consultation — available at appointment time via Zegocloud
- My Prescriptions — prescriptions written by the doctor with medicine details
- Order Medicines — auto-fills cart from prescription and processes payment via Razorpay UPI
- Live vitals sharing toggle for this doctor (built in Phase 3)

Step 4 — Connecting to Multiple Hospitals Simultaneously

A patient can connect to more than one hospital at once (e.g., Apollo for regular checkups, Fortis for a specialist). Each hospital has its own doctor card, chat room, reports section, appointment page, and prescriptions. WebSocket rooms are isolated per patient-doctor pair — data never crosses between hospitals.

6. Clinical Features Unlocked on Connection — Build Notes

Tele-Consultation Chat

Direct human-to-human text messaging between the connected patient and her assigned doctor, opened the moment the access code is used. This is completely separate from the LLM chat (Phase 4) — the doctor only ever sees this human-to-human channel, never the AI conversation.

My Reports

Medical reports (blood test, scan, etc.) uploaded by the hospital and viewable by the patient as PDF. New report uploads trigger a patient notification (detailed in Phase 4).

Book Appointment

Patient views available slots from her connected doctor and books directly. Appointment bookings trigger notifications to both patient and hospital (detailed in Phase 4).

Video Consultation

Available at the scheduled appointment time, delivered via Zegocloud.

My Prescriptions and Order Medicines

Prescriptions are written by the doctor with medicine details and appear under My Prescriptions. "Order Medicines" auto-fills the cart from the prescription and processes payment via Razorpay UPI.

7. Doctor Side — Unlocking Hospital Access and Viewing Patients

Unlocking Hospital Access

To see patients at a hospital, the doctor unlocks it using the hospital-issued credentials (from section 1 above), entered from inside his own personal dashboard — this does not create a second account.

Flow: Doctor opens the hospital's public page from the search bar → clicks "Unlock My Patient Access" → enters the email + password the hospital gave him at registration → credentials verified against hospital records → hospital moves into "My Unlocked Hospitals" permanently → his patient list at that hospital becomes visible.

Edge Cases — Unlocking Access

Edge Case	What Happens
Doctor enters the wrong hospital-issued password	"Incorrect credentials. Please check with your hospital." He may retry.
Doctor tries to unlock a hospital he was never registered at	Verification fails — no matching doctor record to check against.
Hospital deactivates the doctor after he has already unlocked it	Unlock revoked immediately; banner: "Your access at [Hospital] has been deactivated by the hospital." His personal account and other unlocked hospitals are unaffected.
Doctor loses his hospital-issued password after unlocking once	Not needed again — unlock is remembered permanently. Hospital-side credential resets do not force a re-unlock.

Viewing the Patient List and Opening a Patient Record

Once unlocked, that hospital's section of the doctor's dashboard shows the list of patients mapped to him there — exactly the mapping the hospital created. Clicking a patient opens her record showing name, age, gender, and registration date (when she was first mapped to him under this hospital), with a "View Vitals" button leading to Phase 3's live/daily/weekly/monthly views.

Edge Cases — Patient List and Records

Edge Case	What Happens
Doctor has zero patients at a hospital just unlocked	List shows "No patients assigned yet."
Hospital maps a new patient after doctor already unlocked	She appears automatically next time he opens the list — no re-unlock needed.
Hospital reassigns his patient to a different doctor	She disappears from his list immediately; historical chat, reports, and prescriptions remain in his own records for reference.
Patient shared pre-connection self-recorded history with him	Record and vitals include everything from day one, marked source "self" for pre-connection entries (see Phase 3).
Patient's age or gender field left blank at registration	Field displays "Not provided" rather than blank or broken.

8. Ongoing Hospital Management (Post-Registration)

Beyond initial registration, the hospital dashboard supports continuous management:

- View all doctors with a live count of patients mapped to each
- Open any doctor to see his complete patient list
- Reassign a patient from one doctor to another within the same hospital
- Deactivate a doctor, following the rules in Section 1
- Search patients by name or phone number across the entire hospital
- Check any patient's access code status at a glance

Edge Cases — Ongoing Management

Edge Case	What Happens
Hospital deactivates a doctor with pending appointments	Pending appointments auto-flagged "Doctor unavailable — please reschedule." Hospital prompted to reassign before deactivation finalises.
Two staff members edit the same doctor's profile simultaneously	Last save wins — no built-in conflict merge (documented prototype limitation).
Hospital needs separate logins for different staff	Out of scope for this prototype — single shared hospital login for all staff.

9. Multi-Hospital Access for a Doctor

A doctor's personal account is singular, but his patient access is not. If he is visiting at XYZ Hospital and permanent at ABC Hospital, he unlocks each independently, using the separate credentials each hospital issued him.

Hospital	His Role There	Unlock Status	Patients Visible
XYZ Hospital	Visiting	Unlocked with XYZ's credentials	Only XYZ Hospital's mapped patients
ABC Hospital	Permanent	Unlocked with ABC's credentials	Only ABC Hospital's mapped patients

Edge Cases — Multi-Hospital Access

Edge Case	What Happens
Same real doctor registered with slightly different spelling/qualification at two hospitals	No automatic reconciliation. Each hospital's entry is authoritative for its own portal.
Doctor unlocks a hospital, then hospital-side credentials change later	Existing unlock unaffected unless the hospital explicitly deactivates him.
Doctor tries to unlock the same hospital twice	Not necessary — option changes to "View Patients" instead of "Unlock"; no re-entry of credentials.

Phase 2 Summary Table

Component	Key Rule Established
Doctor registration by hospital	Full credentials required; hospital-specific login; independent per hospital for multi-hospital doctors.
Doctor directory	Fully public, read-only, no login needed to browse.
Patient-doctor mapping	Hospital creates mapping + access code only, never the patient's login.
Access code	8-character, 30-day validity, single-use, scoped to one hospital-doctor pairing.

Component	Key Rule Established
Doctor unlock	Hospital-issued credentials unlock a hospital once, permanently — not a second login.
Clinical features unlocked	Tele-consultation chat, My Reports (PDF), Book Appointment, Video Consultation (Zegocloud), My Prescriptions, Order Medicines (Razorpay UPI).
Deactivation	Revokes access instantly; preserves historical records.